2

Estimating the CDF and Statistical Functionals

The first problem we consider is estimating the CDF. By itself, this is not a very interesting problem. However, it is the first step towards solving more important problems such as estimating statistical functionals.

2.1 The CDF

We begin with the problem of estimating a CDF (cumulative distribution function). Let $X_1, \ldots, X_n \sim F$ where $F(x) = \mathbb{P}(X \leq x)$ is a distribution function on the real line. We estimate F with the empirical distribution function.

2.1 Definition. *The* **empirical distribution function** $\widehat{F}_n$ *is the* CDF *that puts mass* $1/n$ *at each data point* X_i*. Formally,*

$$\widehat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} I(X_i \leq x) \tag{2.2}$$

where

$$I(X_i \leq x) = \begin{cases} 1 & \text{if } X_i \leq x \\ 0 & \text{if } X_i > x. \end{cases}$$

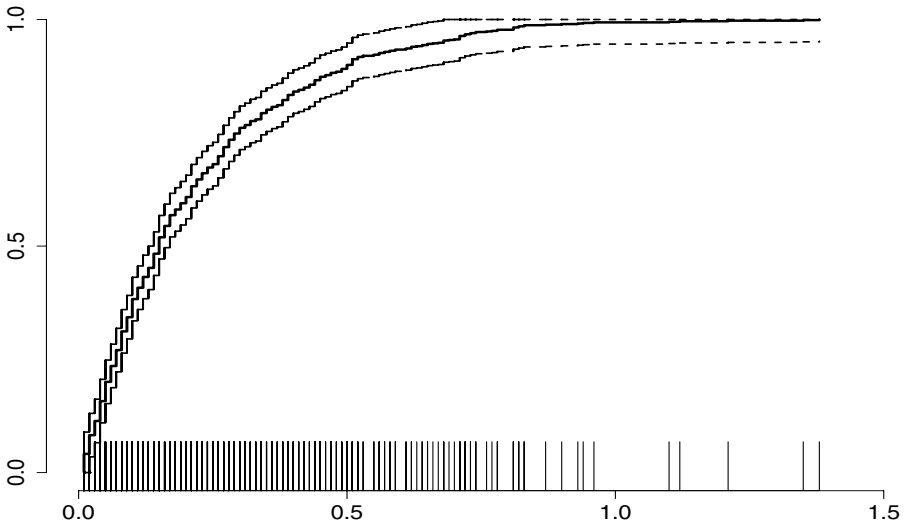


FIGURE 2.1. Nerve data. Each vertical line represents one data point. The solid line is the empirical distribution function. The lines above and below the middle line are a 95 percent confidence band.

2.3 Example (Nerve data)**.** Cox and Lewis (1966) reported 799 waiting times between successive pulses along a nerve fiber. Figure 2.1 shows the data and the empirical CDF $\widehat{F}_n$. ■

The following theorem gives some properties of $\widehat{F}_n(x)$.

2.4 Theorem. *Let $X_1, \ldots, X_n \sim F$ and let $\widehat{F}_n$ be the empirical* CDF*. Then:*

1. *At any fixed value of x,*

$$\mathbb{E}\left(\widehat{F}_n(x)\right) = F(x) \quad \text{and} \quad \mathbb{V}\left(\widehat{F}_n(x)\right) = \frac{F(x)(1-F(x))}{n}.$$

Thus, MSE $= \frac{F(x)(1-F(x))}{n} \to 0$ *and hence* $\widehat{F}_n(x) \xrightarrow{\text{P}} F(x)$.

2. (Glivenko–Cantelli Theorem).

$$\sup_x |\widehat{F}_n(x) - F(x)| \xrightarrow{\text{a.s.}} 0.$$

3. (Dvoretzky–Kiefer–Wolfowitz (DKW) inequality). *For any $\epsilon > 0$,*

$$\mathbb{P}\left(\sup_x |F(x) - \widehat{F}_n(x)| > \epsilon\right) \leq 2e^{-2n\epsilon^2}. \tag{2.5}$$

From the DKW inequality, we can construct a confidence set. Let $\epsilon_n^2 = \log(2/\alpha)/(2n)$, $L(x) = \max\{\widehat{F}_n(x) - \epsilon_n, 0\}$ and $U(x) = \min\{\widehat{F}_n(x) + \epsilon_n, 1\}$. It follows from (2.5) that for any F,

$$\mathbb{P}(L(x) \leq F(x) \leq U(x) \text{ for all } x) \geq 1 - \alpha.$$

Thus, $(L(x), U(x))$ is a nonparametric $1-\alpha$ **confidence band**.[1]
To summarize:

2.6 Theorem. *Let*

$$\begin{aligned} L(x) &= \max\{\widehat{F}_n(x) - \epsilon_n, 0\} \\ U(x) &= \min\{\widehat{F}_n(x) + \epsilon_n, 1\} \end{aligned}$$

where

$$\epsilon_n = \sqrt{\frac{1}{2n}\log\left(\frac{2}{\alpha}\right)}.$$

Then, for all F and all n,

$$\mathbb{P}\Big(L(x) \leq F(x) \leq U(x) \quad \text{for all } x\Big) \geq 1 - \alpha.$$

2.7 Example. The dashed lines in Figure 2.1 give a 95 percent confidence band using $\epsilon_n = \sqrt{\frac{1}{2n}\log\left(\frac{2}{.05}\right)} = .048$. ■

2.2 Estimating Statistical Functionals

A **statistical functional** $T(F)$ is any function of F. Examples are the mean $\mu = \int x\, dF(x)$, the variance $\sigma^2 = \int (x-\mu)^2 dF(x)$ and the median $m = F^{-1}(1/2)$.

2.8 Definition. *The* **plug-in estimator** *of $\theta = T(F)$ is defined by*

$$\widehat{\theta}_n = T(\widehat{F}_n). \tag{2.9}$$

A functional of the form $\int a(x)dF(x)$ is called a **linear functional.** Recall that $\int a(x)dF(x)$ is defined to be $\int a(x)f(x)dx$ in the continuous case

[1] There exist tighter confidence bands but we use the DKW band because it is simple.

and $\sum_j a(x_j)f(x_j)$ in the discrete case. The empirical CDF $\widehat{F}_n(x)$ is discrete, putting mass $1/n$ at each X_i. Hence, if $T(F) = \int a(x)dF(x)$ is a linear functional then we have:

The plug-in estimator for linear functional $T(F) = \int a(x)dF(x)$ is:

$$T(\widehat{F}_n) = \int a(x)d\widehat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} a(X_i). \tag{2.10}$$

Sometimes we can find the estimated standard error $\widehat{\mathsf{se}}$ of $T(\widehat{F}_n)$ by doing some direct calculations. However, in other cases it is not obvious how to estimate the standard error. Later we will discuss methods for finding $\widehat{\mathsf{se}}$. For now, let us just assume that somehow we can find $\widehat{\mathsf{se}}$. In many cases, it turns out that

$$T(\widehat{F}_n) \approx N(T(F), \widehat{\mathsf{se}}^2). \tag{2.11}$$

In that case, an approximate $1-\alpha$ confidence interval for $T(F)$ is then

$$T(\widehat{F}_n) \pm z_{\alpha/2}\,\widehat{\mathsf{se}} \tag{2.12}$$

where z_α is defined by: $\mathbb{P}(Z > z_\alpha) = \alpha$ with $Z \sim N(0,1)$. We will call (2.12) the **Normal-based interval.**

2.13 Example (The mean)**.** Let $\mu = T(F) = \int x\,dF(x)$. The plug-in estimator is $\widehat{\mu} = \int x\,d\widehat{F}_n(x) = \overline{X}_n$. The standard error is $\mathsf{se} = \sqrt{\mathbb{V}(\overline{X}_n)} = \sigma/\sqrt{n}$. If $\widehat{\sigma}$ denotes an estimate of σ, then the estimated standard error is $\widehat{\mathsf{se}} = \widehat{\sigma}/\sqrt{n}$. A Normal-based confidence interval for μ is $\overline{X}_n \pm z_{\alpha/2}\,\widehat{\sigma}/\sqrt{n}$. ∎

2.14 Example (The variance)**.** Let $\sigma^2 = \mathbb{V}(X) = \int x^2\,dF(x) - \left(\int x\,dF(x)\right)^2$. The plug-in estimator is

$$\begin{aligned}
\widehat{\sigma}^2 &= \int x^2 d\widehat{F}_n(x) - \left(\int x d\widehat{F}_n(x)\right)^2 \\
&= \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^2 \\
&= \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2.
\end{aligned}$$

This is different than the usual unbiased sample variance

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2.$$

In practice, there is little difference between $\widehat{\sigma}^2$ and S_n^2. ■

2.15 Example (The skewness). Let μ and σ^2 denote the mean and variance of a random variable X. The skewness — which measures the lack of symmetry of a distribution — is defined to be

$$\kappa = \frac{\mathbb{E}(X-\mu)^3}{\sigma^3} = \frac{\int (x-\mu)^3 dF(x)}{\left\{\int (x-\mu)^2 dF(x)\right\}^{3/2}}.$$

To find the plug-in estimate, first recall that $\widehat{\mu} = n^{-1}\sum_{i=1}^n X_i$ and $\widehat{\sigma}^2 = n^{-1}\sum_{i=1}^n (X_i - \widehat{\mu})^2$. The plug-in estimate of κ is

$$\widehat{\kappa} = \frac{\int (x-\mu)^3 d\widehat{F}_n(x)}{\left\{\int (x-\mu)^2 d\widehat{F}_n(x)\right\}^{3/2}} = \frac{\frac{1}{n}\sum_{i=1}^n (X_i - \widehat{\mu})^3}{\widehat{\sigma}^3}. \quad ■$$

2.16 Example (Correlation). Let $Z = (X, Y)$ and let $\rho = T(F) = \mathbb{E}(X - \mu_X)(Y - \mu_Y)/(\sigma_x \sigma_y)$ denote the correlation between X and Y, where $F(x, y)$ is bivariate. We can write $T(F) = a(T_1(F), T_2(F), T_3(F), T_4(F), T_5(F))$ where

$$\begin{array}{lll} T_1(F) = \int x\, dF(z) & T_2(F) = \int y\, dF(z) & T_3(F) = \int xy\, dF(z) \\ T_4(F) = \int x^2\, dF(z) & T_5(F) = \int y^2\, dF(z) & \end{array}$$

and

$$a(t_1, \ldots, t_5) = \frac{t_3 - t_1 t_2}{\sqrt{(t_4 - t_1^2)(t_5 - t_2^2)}}.$$

Replace F with $\widehat{F}_n$ in $T_1(F)$, ..., $T_5(F)$, and take

$$\widehat{\rho} = a(T_1(\widehat{F}_n), T_2(\widehat{F}_n), T_3(\widehat{F}_n), T_4(\widehat{F}_n), T_5(\widehat{F}_n)).$$

We get

$$\widehat{\rho} = \frac{\sum_{i=1}^n (X_i - \overline{X}_n)(Y_i - \overline{Y}_n)}{\sqrt{\sum_{i=1}^n (X_i - \overline{X}_n)^2}\sqrt{\sum_{i=1}^n (Y_i - \overline{Y}_n)^2}}$$

which is called the **sample correlation**. ■

2.17 Example (Quantiles). Let F be strictly increasing with density f. Let $T(F) = F^{-1}(p)$ be the pth quantile. The estimate of $T(F)$ is $\widehat{F}_n^{-1}(p)$. We have to be a bit careful since $\widehat{F}_n$ is not invertible. To avoid ambiguity we define $\widehat{F}_n^{-1}(p) = \inf\{x : \ \widehat{F}_n(x) \geq p\}$. We call $\widehat{F}_n^{-1}(p)$ the pth **sample quantile**. ■

The Glivenko–Cantelli theorem ensures that $\widehat{F}_n$ converges to F. This suggests that $\widehat{\theta}_n = T(\widehat{F}_n)$ will converge to $\theta = T(F)$. Furthermore, we would hope that under reasonable conditions, $\widehat{\theta}_n$ will be asymptotically normal. This leads us to the next topic.

2.3 Influence Functions

The influence function is used to approximate the standard error of a plug-in estimator. The formal definition is as follows.

2.18 Definition. *The* **Gâteaux derivative** *of T at F in the direction G is defined by*

$$L_F(G) = \lim_{\epsilon \to 0} \frac{T\big((1-\epsilon)F + \epsilon G\big) - T(F)}{\epsilon}. \tag{2.19}$$

If $G = \delta_x$ is a point mass at x then we write $L_F(x) \equiv L_F(\delta_x)$ and we call $L_F(x)$ the **influence function**. *Thus,*

$$L_F(x) = \lim_{\epsilon \to 0} \frac{T\big((1-\epsilon)F + \epsilon \delta_x\big) - T(F)}{\epsilon}. \tag{2.20}$$

The **empirical influence function** *is defined by $\widehat{L}(x) = L_{\widehat{F}_n}(x)$. Thus,*

$$\widehat{L}(x) = \lim_{\epsilon \to 0} \frac{T\big((1-\epsilon)\widehat{F}_n + \epsilon \delta_x\big) - T(\widehat{F}_n)}{\epsilon}. \tag{2.21}$$

Often we drop the subscript F and write $L(x)$ instead of $L_F(x)$.

2.22 Theorem. *Let $T(F) = \int a(x)dF(x)$ be a linear functional. Then:*

1. *$L_F(x) = a(x) - T(F)$ and $\widehat{L}(x) = a(x) - T(\widehat{F}_n)$.*
2. *For any G,*
$$T(G) = T(F) + \int L_F(x)dG(x). \tag{2.23}$$
3. *$\int L_F(x)dF(x) = 0$.*
4. *Let $\tau^2 = \int L_F^2(x)dF(x)$. Then, $\tau^2 = \int (a(x) - T(F))^2 dF(x)$ and if $\tau^2 < \infty$,*
$$\sqrt{n}(T(F) - T(\widehat{F}_n)) \rightsquigarrow N(0, \tau^2). \tag{2.24}$$
5. *Let*
$$\widehat{\tau}^2 = \frac{1}{n}\sum_{i=1}^n \widehat{L}^2(X_i) = \frac{1}{n}\sum_{i=1}^n (a(X_i) - T(\widehat{F}_n))^2. \tag{2.25}$$

Then, $\widehat{\tau}^2 \xrightarrow{\text{P}} \tau^2$ and $\widehat{\mathsf{se}}/\mathsf{se} \xrightarrow{\text{P}} 1$ where $\widehat{\mathsf{se}} = \widehat{\tau}/\sqrt{n}$ and $\mathsf{se} = \sqrt{\mathbb{V}(T(\widehat{F}_n))}$.

6. We have that

$$\frac{\sqrt{n}(T(F) - T(\widehat{F}_n))}{\widehat{\tau}} \rightsquigarrow N(0,1). \tag{2.26}$$

PROOF. The first three claims follow easily from the definition of the influence function. To prove the fourth claim, write

$$\begin{aligned} T(\widehat{F}_n) &= T(F) + \int L_F(x) d\widehat{F}_n(x) \\ &= T(F) + \frac{1}{n}\sum_{i=1}^{n} L_F(X_i). \end{aligned}$$

From the central limit theorem and the fact that $\int L_F(x)dF(x) = 0$, it follows that

$$\sqrt{n}(T(F) - T(\widehat{F}_n)) \rightsquigarrow N(0, \tau^2)$$

where $\tau^2 = \int L_F^2(x)dF(x)$. The fifth claim follows from the law of large numbers. The final statement follows from the fourth and fifth claims and Slutsky's theorem. ■

The theorem above tells us that the influence function $L_F(x)$ behaves like the score function in parametric estimation. To see this, recall that if $f(x;\theta)$ is a parametric model, $\mathcal{L}_n(\theta) = \prod_{i=1}^n f(X_i;\theta)$ is the likelihood function and the maximum likelihood estimator $\widehat{\theta}_n$ is the value of θ that maximizes $\mathcal{L}_n(\theta)$. The score function is $s_\theta(x) = \partial \log f(x;\theta)/\partial\theta$ which, under appropriate regularity conditions, satisfies $\int s_\theta(x) f(x;\theta)dx = 0$ and $\mathbb{V}(\widehat{\theta}_n) \approx \int (s_\theta(x))^2 f(x;\theta)dx/n$. Similarly, for the influence function we have that $\int L_F(x)dF(x) = 0$ and and $\mathbb{V}(T(\widehat{F}_n)) \approx \int L_F^2(x)dF(x)/n$.

If the functional $T(F)$ is not linear, then (2.23) will not hold exactly, but it may hold approximately.

2.27 Theorem. *If T is Hadamard differentiable*[2] *with respect to $d(F,G) = \sup_x |F(x) - G(x)|$ then*

$$\sqrt{n}(T(\widehat{F}_n) - T(F)) \rightsquigarrow N(0, \tau^2) \tag{2.28}$$

where $\tau^2 = \int L_F(x)^2 dF(x)$. Also,

$$\frac{(T(\widehat{F}_n) - T(F))}{\widehat{\mathsf{se}}} \rightsquigarrow N(0,1) \tag{2.29}$$

[2]Hadamard differentiability is defined in the appendix.

where $\widehat{\mathsf{se}} = \widehat{\tau}/\sqrt{n}$ *and*

$$\widehat{\tau} = \frac{1}{n}\sum_{i=1}^{n} L^2(X_i). \tag{2.30}$$

We call the approximation $(T(\widehat{F}_n) - T(F))/\widehat{\mathsf{se}} \approx N(0,1)$ the **nonparametric delta method**. From the normal approximation, a large sample confidence interval is $T(\widehat{F}_n) \pm z_{\alpha/2}\,\widehat{\mathsf{se}}$. This is only a pointwise asymptotic confidence interval. In summary:

The Nonparametric Delta Method

A $1-\alpha$, pointwise asymptotic confidence interval for $T(F)$ is

$$T(\widehat{F}_n) \pm z_{\alpha/2}\,\widehat{\mathsf{se}} \tag{2.31}$$

where

$$\widehat{\mathsf{se}} = \frac{\widehat{\tau}}{\sqrt{n}} \quad \text{and} \quad \widehat{\tau}^2 = \frac{1}{n}\sum_{i=1}^{n} \widehat{L}^2(X_i).$$

2.32 Example (The mean)**.** Let $\theta = T(F) = \int x\, dF(x)$. The plug-in estimator is $\widehat{\theta} = \int x\, d\widehat{F}_n(x) = \overline{X}_n$. Also, $T((1-\epsilon)F + \epsilon\delta_x) = (1-\epsilon)\theta + \epsilon x$. Thus, $L(x) = x - \theta$, $\widehat{L}(x) = x - \overline{X}_n$ and $\widehat{\mathsf{se}}^2 = \widehat{\sigma}^2/n$ where $\widehat{\sigma}^2 = n^{-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2$. A pointwise asymptotic nonparametric 95 percent confidence interval for θ is $\overline{X}_n \pm 2\,\widehat{\mathsf{se}}$. ■

Sometimes statistical functionals take the form $T(F) = a(T_1(F), \ldots, T_m(F))$ for some function $a(t_1, \ldots, t_m)$. By the chain rule, the influence function is

$$L(x) = \sum_{i=1}^{m} \frac{\partial a}{\partial t_i} L_i(x)$$

where

$$L_i(x) = \lim_{\epsilon \to 0} \frac{T_i((1-\epsilon)F + \epsilon\delta_x) - T_i(F)}{\epsilon}. \tag{2.33}$$

2.34 Example (Correlation)**.** Let $Z = (X, Y)$ and let $T(F) = E(X - \mu_X)(Y - \mu_Y)/(\sigma_x\sigma_y)$ denote the correlation where $F(x, y)$ is bivariate. Recall that $T(F) = a(T_1(F), T_2(F), T_3(F), T_4(F), T_5(F))$ where

$$\begin{array}{lll} T_1(F) = \int x\, dF(z) & T_2(F) = \int y\, dF(z) & T_3(F) = \int xy\, dF(z) \\ T_4(F) = \int x^2\, dF(z) & T_5(F) = \int y^2\, dF(z) & \end{array}$$

and

$$a(t_1, \ldots, t_5) = \frac{t_3 - t_1 t_2}{\sqrt{(t_4 - t_1^2)(t_5 - t_2^2)}}.$$

It follows from (2.33) that

$$L(x,y) = \widetilde{x}\widetilde{y} - \frac{1}{2}T(F)(\widetilde{x}^2 + \widetilde{y}^2)$$

where

$$\widetilde{x} = \frac{x - \int x dF}{\sqrt{\int x^2 dF - (\int x dF)^2}}, \quad \widetilde{y} = \frac{y - \int y dF}{\sqrt{\int y^2 dF - (\int y dF)^2}}. \quad \blacksquare$$

2.35 Example (Quantiles). Let F be strictly increasing with positive density f. The $T(F) = F^{-1}(p)$ be the pth quantile. The influence function is (see Exercise 10)

$$L(x) = \begin{cases} \frac{p-1}{f(\theta)}, & x \leq \theta \\ \frac{p}{f(\theta)}, & x > \theta. \end{cases}$$

The asymptotic variance of $T(\widehat{F}_n)$ is

$$\frac{\tau^2}{n} = \frac{1}{n}\int L^2(x) dF(x) = \frac{p(1-p)}{nf^2(\theta)}. \tag{2.36}$$

To estimate this variance we need to estimate the density f. Later we shall see that the bootstrap provides a simpler estimate of the variance. ■

2.4 Empirical Probability Distributions

This section discusses a generalization of the DKW inequality. The reader may skip this section if desired. Using the empirical CDF to estimate the true CDF is a special case of a more general idea. Let $X_1, \ldots, X_n \sim P$ be an IID sample from a probability measure P. Define the **empirical probability distribution** $\mathbb{P}_n$ by

$$\widehat{\mathbb{P}}_n(A) = \frac{\text{number of } X_i \in A}{n}. \tag{2.37}$$

We would like to be able to say that $\mathbb{P}_n$ is close to P in some sense. For a fixed A we know that $n\widehat{P}_n(A) \sim \text{Binomial}(n,p)$ where $p = P(A)$. By Hoeffding's inequality, it follows that

$$\mathbb{P}\big(|\widehat{P}_n(A) - P(A)| > \epsilon\big) \leq 2e^{-2n\epsilon^2}. \tag{2.38}$$

We would like to extend this to be a statement of the form

$$\mathbb{P}\left(\sup_{A\in\mathcal{A}} |\widehat{P}_n(A) - P(A)| > \epsilon\right) \leq \text{something small}$$

for some class of sets $\mathcal{A}$. This is exactly what the DKW inequality does by taking $\mathcal{A} = \{A = (-\infty, t] : \ t \in \mathbb{R}\}$. But DKW is only useful for one-dimensional random variables. We can get a more general inequality by using Vapnik–Chervonenkis (VC) theory.

Let $\mathcal{A}$ be a class of sets. Given a finite set $R = \{x_1, \ldots, x_n\}$ let

$$N_{\mathcal{A}}(R) = \#\left\{ R \bigcap A : \ A \in \mathcal{A} \right\} \tag{2.39}$$

be the number of subsets of R "picked out" as A varies over $\mathcal{A}$. We say that R is **shattered** by $\mathcal{A}$ if $N_{\mathcal{A}}(R) = 2^n$. The **shatter coefficient** is defined by

$$s(\mathcal{A}, n) = \max_{R \in \mathcal{F}_n} N_{\mathcal{A}}(R) \tag{2.40}$$

where $\mathcal{F}_n$ consists of all finite sets of size n.

2.41 Theorem (Vapnik and Chervonenkis, 1971)**.** *For any P, n and $\epsilon > 0$,*

$$\mathbb{P}\left(\sup_{A \in \mathcal{A}} |\widehat{P}_n(A) - P(A)| > \epsilon \right) \leq 8 s(\mathcal{A}, n) e^{-n\epsilon^2/32}. \tag{2.42}$$

Theorem 2.41 is only useful if the shatter coefficients do not grow too quickly with n. This is where VC dimension enters. If $s(\mathcal{A}, n) = 2^n$ for all n set $\mathrm{VC}(\mathcal{A}) = \infty$. Otherwise, define $\mathrm{VC}(\mathcal{A})$ to be the largest k for which $s(\mathcal{A}, k) = 2^k$. We call $\mathrm{VC}(\mathcal{A})$ the **Vapnik–Chervonenkis** dimension of $\mathcal{A}$. Thus, the VC-dimension is the size of the largest finite set F that is shattered by $\mathcal{A}$. The following theorem shows that if $\mathcal{A}$ has finite VC-dimension then the shatter coefficients grow as a polynomial in n.

2.43 Theorem. *If $\mathcal{A}$ has finite VC-dimension v, then*

$$s(\mathcal{A}, n) \leq n^v + 1.$$

In this case,

$$\mathbb{P}\left(\sup_{A \in \mathcal{A}} |\widehat{P}_n(A) - P(A)| > \epsilon \right) \leq 8(n^v + 1) e^{-n\epsilon^2/32}. \tag{2.44}$$

2.45 Example. Let $\mathcal{A} = \{(-\infty, x];\ x \in \mathbb{R}\}$. Then $\mathcal{A}$ shatters every one point set $\{x\}$ but it shatters no set of the form $\{x, y\}$. Therefore, $\mathrm{VC}(\mathcal{A}) = 1$. Since, $\mathbb{P}((-\infty, x]) = F(x)$ is the CDF and $\widehat{\mathbb{P}}_n((-\infty, x]) = \widehat{F}_n(x)$ is the empirical CDF, we conclude that

$$\mathbb{P}\left(\sup_x |\widehat{F}_n(x) - F(x)| > \epsilon \right) \leq 8(n+1) e^{-n\epsilon^2/32}$$

which is looser than the DKW bound. This shows that the bound (2.42) is not the tightest possible. ∎

2.46 Example. Let $\mathcal{A}$ be the set of closed intervals on the real line. Then $\mathcal{A}$ shatters $S = \{x, y\}$ but it cannot shatter sets with three points. Consider $S = \{x, y, z\}$ where $x < y < z$. One cannot find an interval A such that $A \bigcap S = \{x, z\}$. So, $\text{VC}(\mathcal{A}) = 2$. ∎

2.47 Example. Let $\mathcal{A}$ be all linear half-spaces on the plane. Any three-point set (not all on a line) can be shattered. No four-point set can be shattered. Consider, for example, four points forming a diamond. Let T be the leftmost and rightmost points. This set cannot be picked out. Other configurations can also be seen to be unshatterable. So $\text{VC}(\mathcal{A}) = 3$. In general, halfspaces in $\mathcal{R}^d$ have VC dimension $d + 1$. ∎

2.48 Example. Let $\mathcal{A}$ be all rectangles on the plane with sides parallel to the axes. Any four-point set can be shattered. Let S be a five-point set. There is one point that is not leftmost, rightmost, uppermost or lowermost. Let T be all points in S except this point. Then T can't be picked out. So, we have that $\text{VC}(\mathcal{A}) = 4$. ∎

2.5 Bibliographic Remarks

Further details on statistical functionals can be found in Serfling (1980), Davison and Hinkley (1997), Shao and Tu (1995), Fernholz (1983) and van der Vaart (1998). Vapnik–Chervonenkis theory is discussed in Devroye et al. (1996), van der Vaart (1998) and van der Vaart and Wellner (1996).

2.6 Appendix

Here are some details about Theorem 2.27. Let $\mathfrak{F}$ denote all distribution functions and let $\mathcal{D}$ denote the linear space generated by $\mathfrak{F}$. Write $T((1-\epsilon)F + \epsilon G) = T(F + \epsilon D)$ where $D = G - F \in \mathcal{D}$. The Gateâux derivative, which we now write as $L_F(D)$, is defined by

$$\lim_{\epsilon \downarrow 0} \left| \frac{T(F + \epsilon D) - T(F)}{\epsilon} - L_F(D) \right| \to 0.$$

Thus $T(F + \epsilon D) \approx \epsilon L_F(D) + o(\epsilon)$ and the error term $o(\epsilon)$ goes to 0 as $\epsilon \to 0$. Hadamard differentiability requires that this error term be small uniformly over compact sets. Equip $\mathcal{D}$ with a metric d. T is **Hadamard differentiable**

at F if there exists a linear functional L_F on $\mathcal{D}$ such that for any $\epsilon_n \to 0$ and $\{D, D_1, D_2, \ldots\} \subset \mathcal{D}$ such that $d(D_n, D) \to 0$ and $F + \epsilon_n D_n \in \mathcal{F}$,

$$\lim_{n\to\infty} \left(\frac{T(F + \epsilon_n D_n) - T(F)}{\epsilon_n} - L_F(D_n) \right) = 0.$$

2.7 Exercises

1. Fill in the details of the proof of Theorem 2.22.

2. Prove Theorem 2.4.

3. (Computer experiment.) Generate 100 observations from a N(0,1) distribution. Compute a 95 percent confidence band for the CDF F. Repeat this 1000 times and see how often the confidence band contains the true distribution function. Repeat using data from a Cauchy distribution.

4. Let $X_1, \ldots, X_n \sim F$ and let $\widehat{F}_n(x)$ be the empirical distribution function. For a fixed x, find the limiting distribution of $\sqrt{\widehat{F}_n(x)}$.

5. Suppose that
$$|T(F) - T(G)| \leq C \, ||F - G||_\infty \tag{2.49}$$
for some constant $0 < C < \infty$ where $||F - G||_\infty = \sup_x |F(x) - G(x)|$. Prove that $T(\widehat{F}_n) \xrightarrow{\text{a.s.}} T(F)$. Suppose that $|X| \leq M < \infty$. Show that $T(F) = \int x \, dF(x)$ satisfies (2.49).

6. Let x and y be two distinct points. Find $\text{Cov}(\widehat{F}_n(x), \widehat{F}_n(y))$.

7. Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ and let $Y_1, \ldots, Y_m \sim \text{Bernoulli}(q)$. Find the plug-in estimator and estimated standard error for p. Find an approximate 90 percent confidence interval for p. Find the plug-in estimator and estimated standard error for $p - q$. Find an approximate 90 percent confidence interval for $p - q$.

8. Let $X_1, \ldots, X_n \sim F$ and let $\widehat{F}$ be the empirical distribution function. Let $a < b$ be fixed numbers and define $\theta = T(F) = F(b) - F(a)$. Let $\widehat{\theta} = T(\widehat{F}_n) = \widehat{F}_n(b) - \widehat{F}_n(a)$. Find the influence function. Find the estimated standard error of $\widehat{\theta}$. Find an expression for an approximate $1 - \alpha$ confidence interval for θ.

9. Verify the formula for the influence function in Example 2.34.

10. Verify the formula for the influence function in Example 2.35. *Hint*: Let $F_\epsilon(y) = (1-\epsilon)F(y) + \epsilon\delta_x(y)$ where $\delta_x(y)$ is a point mass at x, i.e., $\delta_x(y) = 0$ if $y < x$ and $\delta_x(y) = 1$ if $y \geq x$. By the definition of $T(F)$, we have that $p = F_\epsilon(T(F_\epsilon))$. Now differentiate with respect to ϵ and evaluate the derivative at $\epsilon = 0$.

11. Data on the magnitudes of earthquakes near Fiji are available on the book website. Estimate the CDF $F(x)$. Compute and plot a 95 percent confidence envelope for F. Find an approximate 95 percent confidence interval for $F(4.9) - F(4.3)$.

12. Get the data on eruption times and waiting times between eruptions of the Old Faithful geyser from the book website. Estimate the mean waiting time and give a standard error for the estimate. Also, give a 90 percent confidence interval for the mean waiting time. Now estimate the median waiting time. In the next chapter we will see how to get the standard error for the median.

13. In 1975, an experiment was conducted to see if cloud seeding produced rainfall. 26 clouds were seeded with silver nitrate and 26 were not. The decision to seed or not was made at random. Get the data from

    `http://lib.stat.cmu.edu/DASL/Stories/CloudSeeding.html`

    Let $\theta = T(F_1) - T(F_2)$ be the difference in the median precipitation from the two groups. Estimate θ. Estimate the standard error of the estimate and produce a 95 percent confidence interval. To estimate the standard error you will need to use formula (2.36). This formula requires the density f so you will need to insert an estimate of f. What will you do? Be creative.

14. Let $\mathcal{A}$ be the set of two-dimensional spheres. That is, $A \in \mathcal{A}$ if $A = \{(x,y) : (x-a)^2 + (y-b)^2 \leq c^2\}$ for some a, b, c. Find the VC dimension of $\mathcal{A}$.

15. The empirical CDF can be regarded as a nonparametric maximum likelihood estimator. For example, consider data $X_1, \ldots, X_n$ on [0,1]. Divide the interval into bins of width Δ and find the MLE over all distributions with constant density over the bins. Show that the resulting CDF converges to the empirical CDF as $\Delta \to 0$.